

SALES PERFORMANCE ANALYTICS

SQL + Power BI Portfolio Project Documentation

Project Focus	Sales, profitability, customer, product, and regional performance analysis
Primary Tools	SQL, Power BI, Power Query
Database	sales_analysis database - sales table
Output	Interactive sales dashboard and portfolio case study

1. Project Overview

This project analyzes transactional sales data to understand revenue performance, profitability, customers, products, categories, and regions. The workflow combines SQL for structured analysis with Power BI for interactive reporting and business communication.

Business Goal	Give stakeholders a clear view of sales performance and the areas driving revenue and profit.
Analytical Goal	Compare sales and profit across time, categories, regions, customers, and products.
Database	sales_analysis
Main Table	sales

Core dashboard sections

- Date and region filters for interactive exploration.
- KPI cards for Total Sales, Profit, Quantity, and Average Sales.
- Sales trend by month/year.
- Category performance and total sales by region.
- Top 5 customers by sales.

2. Data Structure and Preparation

The analysis uses the existing sales table in the sales_analysis database. Key fields include order dates, shipment information, customer details, geographic fields, product attributes, and financial measures.

Important fields used in the analysis

Date	`Order Date`, `Ship Date`
Customer	`Customer ID`, `Customer Name`, `Segment`
Geography	`Country/Region`, `City`, `State`, `Region`
Product	`Product ID`, `Category`, `Sub-Category`, `Product Name`
Measures	`Sales`, `Quantity`, `Discount`, `Profit`

Preparation workflow

- Validated column names and data types before analysis.
- Checked the date field for proper time-based grouping.
- Reviewed sales, quantity, discount, and profit fields for usable numeric values.
- Used aggregation in SQL for business summaries rather than working directly from raw transaction rows in the dashboard.
- Connected prepared SQL views / query outputs to Power BI for reporting.

3. Business Analysis and SQL Workflow

The project is built around business questions. The analysis moves from descriptive performance to deeper investigation.

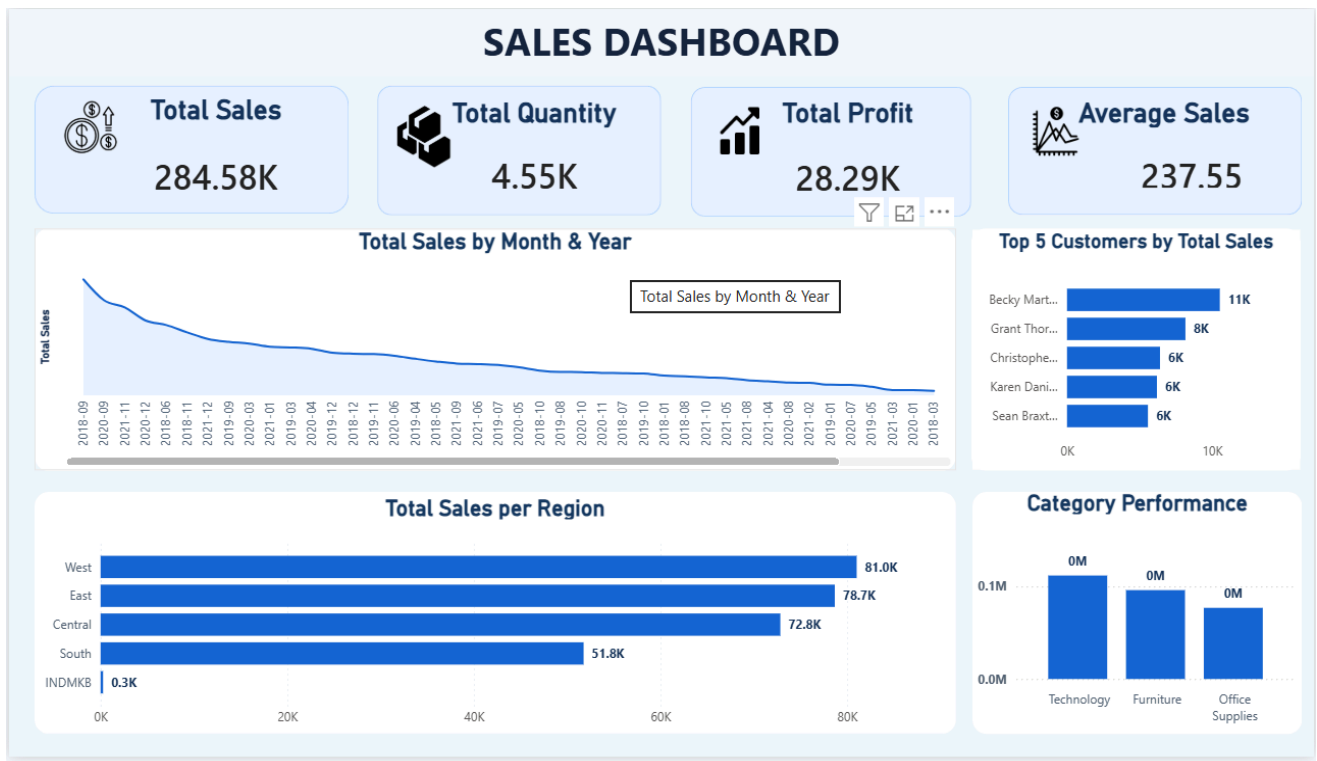
Regional Performance	Compare total sales and total profit by region.
Customer Value	Identify top customers based on total sales and review their profit contribution.
Category Performance	Compare category-level sales and profit to understand where revenue and profit come from.
Discount Investigation	Explore average discount alongside sales and profit to identify areas that may deserve further review.
Monthly Trend	Compare current and previous month sales to identify growth and decline periods.

Representative SQL pattern - monthly comparison

```
WITH monthly_sales AS ( SELECT DATE_FORMAT('Order Date', '%Y-%m') AS Month, SUM(Sales) AS Total_Sales FROM sales GROUP BY DATE_FORMAT('Order Date', '%Y-%m') ) SELECT Month, Total_Sales, LAG(Total_Sales) OVER (ORDER BY Month) AS Previous_Month_Sales, Total_Sales - LAG(Total_Sales) OVER (ORDER BY Month) AS Sales_Change FROM monthly_sales ORDER BY Month;
```

The analysis also uses grouped summaries, HAVING filters, subqueries, CTEs, and views depending on the business question.

4. Dashboard Screenshot



5. Key Insights and Interpretation

The structure below is intentionally ready for the analyst to customize after inserting the final dashboard screenshot and confirming the latest figures.

Revenue Drivers	Identify the regions, categories, and customers contributing the largest share of total sales.
Profit Drivers	Compare profit with sales so high revenue does not automatically get treated as high profitability.
Customer Concentration	Review the top customers and whether their profit contribution is consistent with their sales volume.
Product Performance	Use category and product views to identify strong performers and weaker areas that may require investigation.

Trend Changes	Use month-over-month changes to highlight periods of growth or decline and prompt follow-up analysis.
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6. Recommendations

- Monitor regional and category performance regularly using the dashboard filters.
- Investigate high-sales, low-profit segments before making pricing or discount decisions.
- Review the customer mix and concentration of sales to support account planning.
- Use monthly comparisons to identify periods that require deeper operational or commercial investigation.

7. Technical Summary

Database	sales_analysis
Table	sales
SQL Environment	MySQL / MariaDB-compatible environment
BI Tool	Microsoft Power BI
Analysis Techniques	GROUP BY, HAVING, subqueries, CTEs, window functions, LAG, views
Visualization	KPI cards, trend line, category analysis, region analysis, top-customer view
Portfolio Use	Project card thumbnail + linked PDF case study

Conclusion

This project demonstrates an end-to-end sales analytics workflow: start with a transactional database, translate business questions into SQL analysis, prepare summarized outputs, build an interactive Power BI dashboard, and communicate business-relevant insights. The final case study should emphasize the reasoning behind the analysis, not only the visuals or SQL syntax.